

Saumya Patel

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EDUCATION

University of Alberta

Bachelor of Science in Computing Science (Specialization) | GPA: 3.71

Expected Graduation – May 2028

Edmonton, AB

- **Relevant Coursework:** Software Engineering, Algorithms & Data Structures, Database Management Systems, Operating System Concepts, Parallel & Distributed Systems, Machine Learning, Computer Architecture, Compiler Design

EXPERIENCE

Software Engineer

Aug 2026 – Present

BuildSouk (B2B E-Commerce Marketplace)

Dubai, UAE

- Architected an enterprise B2B marketplace backend from scratch in **TypeScript**, **Next.js**, and **PostgreSQL 16**, enforcing multi-tenant isolation via a 4-role connection pool and database-level **FORCE Row-Level Security (RLS)**.
- Engineered a bilateral **Order Finite State Machine** and zero-float financial engine using **BigInt** micro-fils arithmetic, snapshotting immutable legal contracts (LPOs/GRNs) with line-item partial acceptance and dispute workflows.
- Authored an **8,600-line CI Schema Guard** asserting against **pg_catalog** to structurally block permission regressions, pairing with an automated **229-test Playwright E2E harness** and parallel isolated DB runners in **CI/CD**.
- Integrated an automated **Firecrawl** data ingestion pipeline normalizing distributor catalogs into canonical SKUs with dual-unit conversions, backed by **Redis** sliding-window rate limiting to prevent abuse.

Teaching Assistant | CMPUT 201 (Practical Programming Methodology)

Sep 2025 – Dec 2025

University of Alberta

Edmonton, AB

- Mentored **50+ students** in practical **C programming**, memory safety, and **GDB** debugging, conducting weekly laboratory sessions and peer **code reviews** across **Git** and standard build toolchains (**Make/GCC**).

Research Assistant

May 2025 – Sep 2025

Compiler Design and Optimization Lab

Edmonton, AB

- Researched compiler optimization and pipelined **RISC-V architecture** under Prof. J. Nelson Amaral, developing automated validation harnesses to benchmark runtime efficiency, register allocation, and binary execution.

Software Development Intern

Jun 2024 – Aug 2024

CubeZix Technologies

Dubai, UAE

- Built automated CI/CD and deployment pipelines on **Microsoft Azure** with third-party **REST API** integrations, eliminating manual deployment steps and **reducing operational turnaround time by 30%**.
- Engineered automated business workflows and Enterprise Resource Planning (**ERP**) modules in **Python** and **JavaScript** (Frappe), streamlining data synchronization across services and **cutting manual processing by 30%**.

PROJECTS

Nivesh – AI Equity Analysis Platform | *Python, FastAPI, Docker, PostgreSQL*

May 2026 – Present

- Architected a full-stack **AI equity analysis platform** using **Docker** and **FastAPI**, deploying **95+ RESTful API endpoints** across 20 modular routers for automated stock screening, valuation modeling, and filings ingestion.
- Engineered an automated **423-test pytest suite** (unit and integration) integrated into a **CI/CD pipeline** achieving a **90.5% pass rate**, enforcing shift-left validation and regression-free API contracts.
- Integrated **Retrieval-Augmented Generation (RAG)** with **ChromaDB** and **PostgreSQL** for semantic search on corporate earnings, orchestrating async **Celery/Redis** worker jobs to accelerate feature delivery.

Gazprea Compiler | *C++, LLVM, MLIR, CMake*

Sep 2025 – Jan 2026

- Engineered an optimizing compiler in **C++** lowering abstract syntax trees through **LLVM** and **MLIR**, collaborating in an **Agile** team using **Jira** for sprint tracking and **Git** for peer code reviews.
- Authored an automated regression test harness executing **1,500+ test cases** across syntax, semantic, and runtime validation stages, **ranking 1st in class** for end-to-end pipeline correctness.

TECHNICAL SKILLS

Languages: Python (AsyncIO), C/C++ (C++17/20), Java, SQL (PostgreSQL, MySQL), JavaScript, TypeScript, Bash, Swift
Frameworks & Libraries: FastAPI, Celery Distributed Workers, Redis, ChromaDB, React, Next.js, Node.js, Tailwind CSS
Cloud & Infrastructure: Docker Containers, Kubernetes, Microsoft Azure, CI/CD Pipelines, Linux/UNIX, Git, Jira
Architecture: Distributed Systems, RESTful APIs, Microservices, Automated Testing (pytest), Structured Logging, TCP/IP
Developer Tools & AI: RAG Systems, AI Development (GitHub Copilot, Cursor), GDB Debugger, Valgrind, CMake, Make/GCC